Manoj Reddy Manchala

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Education

University of California San Diego **Sep 2023 – Jun 2025**
- Master of Science in Mechanical Engineering (Specialized in **Robotics**)
- **Key Courses:** Planning & Learning Robotics, Sensing & Estimation in Robotics, Safety for Autonomous Systems

Indian Institute of Technology, Bombay **Jul 2017 – May 2021**
- Bachelor of Technology in Mechanical Engineering

Experience

Graduate Student Researcher, Existential Robotics Lab (Prof. Nikolay Atanasov) **Mar 2024 – Oct 2025**

Vision Guided Autonomous Navigation:
- Trained deep RL policies (PPO, SAC) that drive an Ackermann-steered car through cluttered, obstacle-rich environments from onboard depth sensing, and deployed them on a real F1TENTH car.
- Built a custom physics simulation of the car and its environments in MuJoCo/DM Control to train them.
- Ran the policies onboard the car's NVIDIA Jetson TX2 by converting them from PyTorch to ONNX and TensorRT.

Cross-Modal Transformer for Temporal Context:
- Architected a cross-modal Vision Transformer integrating CNN-based visual patch embeddings & MLP-encoded state tokens within a Multi-head Transformer Encoder to generate temporally aware state representations for RL policies.
- Outperformed the GNN with **20%** increase in mean velocity, **135%** increase in average coverage, & a **27%** reduction in collision frequency - attributed to Transformer's ability to capture spatiotemporal dependencies.

GNNs for Enhanced Spatial Understanding:
- Implemented a graph neural network (GNN)-based feature extractor, modeling point clouds as structured graphs to enhance 3D object classification and spatial reasoning, significantly improving scene understanding.
- Achieved a **21%** increase in navigation speed, **85%** increase in distance covered & **17%** reduction in collisions.

Tech Consultant, FarmInsta & Growero Technologies **Apr 2026 – Aug 2026**
- Built and deployed an HR and payroll app for Growero, used by its employees: Indian statutory payroll (PF, PT, ESI, TDS), geofenced GPS attendance, and leave and expense approvals (Next.js, TypeScript, PostgreSQL).
- Built and deployed a chatbot for FarmInsta that lets field teams ask about farmer visits and team activity in plain language, using Gemini to write SQL over a read-only copy of the operational database (Python).

Projects

Localization and Mapping:
- Developed a Visual-Inertial SLAM system using an Extended Kalman Filter (EKF), integrating IMU prediction & stereo camera updates to estimate a car's trajectory & 3D landmarks while correcting IMU drift.
- Built a LiDAR-based SLAM system for a differential-drive robot, registering point clouds with Iterative Closest Point & reducing drift with GTSAM pose-graph optimization to generate occupancy & texture maps.

Motion Planning:
- Formulated a dynamic programming approach for a 'Door & Key' navigation, formulating the task as a Markov Decision Process, demonstrating effective path planning for both known and random maps.
- Implemented weighted A* on a 26-connected 3D grid with exact segment–box collision checking, planning collision-free paths through all 7 benchmark environments, up to **29%** shorter than sampling-based RRT.
- Built a receding-horizon MPC (certainty-equivalent control, CasADi/IPOPT) that tracks a path past obstacles under motion noise with **0.21 m** mean error, **4.4**× below a P-controller baseline, and 1 obstacle contact in 4,800 steps vs. 184.

Comma.ai Calibration Challenge (team of 3):
- Trained a Conv-LSTM on dashcam frame sequences stacked with dense optical flow (5-channel input) to estimate camera pitch and yaw, using a turn-weighted MSE because ∼80% of the driving is straight.
- Ablations showed RGB alone gave near-constant predictions and optical flow alone was noisy; fusing both captured the trend in direction of travel, from just five labelled one-minute videos.

Mobile Manipulator:
- Developed a control system for a youBot's autonomous pick & place task in MATLAB & CoppeliaSim, implementing kinematics simulation, trajectory generation & feedforward + PI control that corrects a **30°**, **0.2 m** initial error.

Technical Skills

Robotics	Reinforcement Learning, SLAM, State Estimation, Sensor Fusion, Motion Planning, MPC
Machine Learning	PyTorch, JAX, TensorFlow, Computer Vision, OpenCV, CUDA, ONNX, TensorRT, Jetson
Software	Python, MATLAB, SQL, ROS, GTSAM, MuJoCo, CoppeliaSim, Simulink, Linux, Git